UNIVERSITE DE BOUMERDES UNIVERSITY OF BOUMERDES	*IGEE* *University M'Hamed* *Bougara of Boumerdès*	***Last Name:*** ..... ***First Name:*** ..... ***Group:***	***Grade :***

EE321: Computer Architecture
Control Exam Solution (1h30)
2021 – 2022

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Basic Concepts (8 pts)

1.1 Consider a CPU running at 2 GHz with 200 instructions, 100 instructions are executed in 4 clock cycles, 60 instructions in 7 clock cycles and 40 instructions in 10 clock cycles. What is the speed (MIPS) of the CPU?

*CPI = 4*0,,5 + 7*0,3 + 10*0,2 = 6,1 clock cycles*
Speed = $2*10^9/(6,1*10^6)$ = 328 MIPS *(1 pt)*

1.2 Give the functional table of an active low 3-state buffer.

enable	*in*	*out*
0	*0*	*0*
0	*1*	*1*
1	*0*	*Z*
1	*1*	*Z*

(1 pt)

1.3 Give one application of the 3-state buffer.

It is used to connect registers outputs to the internal CPU data bus *(1 pt)*

1.4 Consider 2's complement representation using 20 bits, complete the following equalities:

a) $(+15)_{10}$ = $(0000FH)_{2'C}$ *$(-15)_{10} = (FFFF1H)_{2'C}$*
b) $(00F00H)_{2'C} = (3840)_{10}$ *$(FF100H)_{2'C} = (-3840)_{10}$* *(1 pt)*

1.5 Consider a 16-bit CPU and the HEX operation: 00F0H-00FFH. Which of the flags: carry, overflow, sign, and zero are set by this operation. Justify.

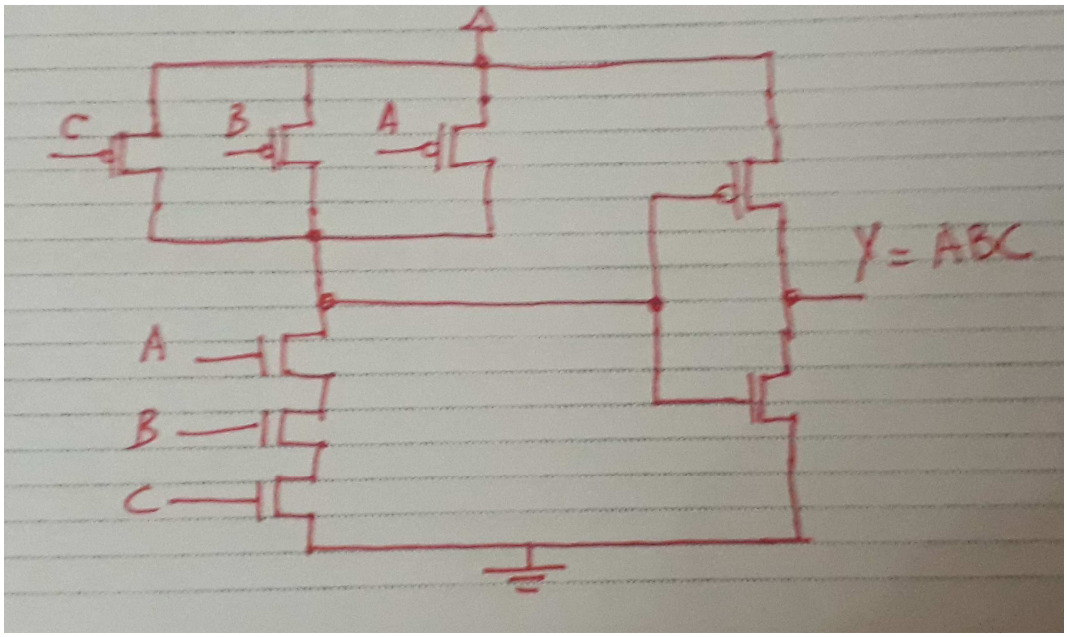
00F0H-00FF = 00F0+FF01=FFF1H
The affected flags are: carry and sign *(1 pt)*

1.6 Give the main advantage and disadvantage of the CMOS technology compared to NMOS technology.

Advantage: Power consumption
Disadvantage: Number of transistors (size) *(1 pt)*

1.7 Draw the transistor schematic of the CMOS gate that implements: $Y = \overline{\overline{ABCD}.\overline{ABC}}$.

$Y = ABCD + ABC = ABC$



(2 pts)

2. CPU Function and Program Execution (12 pts)

Consider a Little-Endian simple 8-bit CPU with the following specifications: 8-bit address bus, memory is byte addressable, instructions opcodes are given in Table 1, binary codes of the general purpose registers (GPR) are given in Table 2, and the instructions formats are given in Table 3 and Table 4. After the fetch cycle of the first instruction of the program, we have: PC=96H, MBR=D0H, and IR=D080H, and after the fetch cycle of the last instruction, we have PC=B6H, MBR=F8H and IR=E0F8H.

2.1 Give the size of the registers : PC, MAR, IR, and MBR. Justify.

The size of registers PC, MAR, and MBR is 8 bits
The size of the register IR is 16 bits *(1 pt)*

2.2 Give the instructions that are using indirect access memory.

LD Rd, # Rs (Opcode=5H)
AD Rd, # Rs (Opcode=CH) *(1 pt)*

2.3 Give the HEX code of the first and last instruction. Justify

The HEX code of the first instruction is: D0H (1-Byte instruction)
The HEX code of the last instruction is: E0F8H (2-Bytes instruction) *(1 pt)*

2.4 Give the content of the registers R0, PC, MAR, MBR, and IR after executing the first instruction.

The HEX code of the first instruction is: D0H
The first instruction is: XOR R0, R0
R0 = 00H (R0 xor R0)
PC = 96H
MAR =95H
MBR =D0H
IR =D080H *(2 pts)*

2.5 Give the content of the registers R0, PC, MAR, MBR, and IR after executing the last instruction.

The HEX code of the last instruction is: E0F8H
The last instruction is: JP F8H (i.e. JP -8)
R0 = XXH
PC = B4 – 8 = ACH
MAR =B5H
MBR =F8H
IR =E0F8H *(2 pts)*

2.6 What is the size of the program? Justify.

Address of the first instruction: 95H = 149
Address of the last byte of the last instruction: B5H = 181
We have 33 (181-149) locations, 1 Byte per location
Size = 33 Bytes = 264 bits *(2 pts)*

2.7 Using register R0 and R1, write an optimized assembly program to add the value -8 to the content of memory location 80H.

```
MV  R1, 80H        ; R1 = address 80H
MV R0, F8H         ; R1 = -8
AD R0, [R1]        ; R0 = (-8) + [R1] = (-8) + [80H]
ST R1, R0          ; Store R0 into memory location 80H
HL                 ; Halt
```
(2 pts)

2.8 Give the HEX code and the size of your program.

HEX code: 118010F8A174F0H
Size: 7 Bytes = 56 bits *(2 pts)*

Instruction	Opcode (hex)	Description
MV Rd, X	1H	Rd = X
MV Rd, Rs	2H	Rd = Rs
LD Rd, Rs	3H	Rd = [Rs]
LD Rd, [X]	4H	Rd = [X]
LD Rd, # Rs	5H	Rd =[[Rs]]
ST Rd, X	6H	[Rd] = X
ST Rd, Rs	7H	[Rd] = Rs
AD Rd, X	8H	Rd = Rd+X
AD Rd, Rs	9H	Rd = Rd+Rs
AD Rd, [Rs]	AH	Rd = Rd+[Rs]
AD Rd, [X]	BH	Rd =Rd+ [X]
AD Rd, # X	CH	Rd=Rd+[[X]]
XOR Rd, Rs	DH	Rd=Rd xor Rs
JP X	EH	PC = PC+X X= 8-bit 2's complement displacement
HL	FH	Halt

Rd = Source/destination register (it can be any GPR).
Rs = Source register (it can be any GPR)
X = 8-bit value/address
R0 = Register R0
[X] = content of memory location X.

Table 1: Instructions description.

Register	Code
R0	00
R1	01
R2	10
R3	11

Table 2: Binary codes of GPRs.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Opcode				0	0	Rd		Operand reference X							

Table 3: 2-Bytes instructions format (for the instructions using operand reference X).

7	6	5	4	3	2	1	0
Opcode				Destination register (Rd)		Source register (Rs)	

Table 4: 1-Byte instructions format (for the instructions without the operand reference X).